Reduced-Bias Estimation for ML, FIML, and Moment-Quadratic SEM

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Purpose

This note records the common estimating-equation convention used by the `estimate::frontier::rbm_*` implementation. The first implementation covered complete-data ML and FIML. The extension adds continuous ULS/GLS/WLS, all-ordinal LS, mixed ordinal LS, and ML2S. All paths use the same linear-constraint-reduced trace algebra; they differ only in how they supply the observed bread and the empirical estimating-function meat.

Objective scale

Let the estimator minimize a per-observation objective

$$f_N(\theta) = \frac{1}{N}\sum_{i=1}^{N} f_i(\theta), \qquad U_N(\theta) = \frac{\partial f_N(\theta)}{\partial \theta}.$$

For likelihood estimators f_i is the normal deviance contribution up to the usual SEM scale. For moment-quadratic estimators

$$f_N(\theta) = \frac{1}{2}\sum_{b} \frac{n_b}{N} d_b(\theta)^{\mathsf{T}} W_b d_b(\theta), \qquad d_b(\theta) = m_b(\theta) - s_b.$$

The implementation stores the observed bread on the total-N scale,

$$J_N(\theta) = N\,\frac{\partial U_N(\theta)}{\partial \theta^{\mathsf{T}}},$$

and the meat as the empirical sum

$$E_N(\theta) = \sum_{i=1}^{N} \psi_i(\theta)\psi_i(\theta)^{\mathsf{T}}, \qquad \sum_{i} \psi_i(\widehat{\theta}) \approx 0.$$

With this convention the trace term is dimensionless:

$$T(\theta) = \mathrm{tr}\{J_N(\theta)^{-1} E_N(\theta)\}.$$

Linear constraints

The lavaanified model may contain linear equality constraints. Write the free-parameter vector as

$$\theta = \theta_0 + K\alpha,$$

where the columns of K span the feasible tangent space. RBM is carried out in α-space:

$$J_\alpha = K^\mathsf{T} J_N K, \qquad E_\alpha = K^\mathsf{T} E_N K, \qquad T_\alpha = \mathrm{tr}(J_\alpha^{-1} E_\alpha).$$

The explicit one-step correction computes the gradient of

$$P(\alpha) = -\frac{1}{2} T_\alpha(\alpha)$$

at the unadjusted solution and applies

$$\Delta_\alpha = J_\alpha^{-1} \frac{\partial P(\alpha)}{\partial \alpha}, \qquad \hat{\theta}_{\mathrm{eRBM}} = \hat{\theta} + K\Delta_\alpha.$$

The implicit estimator minimizes the adjusted objective

$$f_N(\theta) - \frac{1}{N} P(\theta) = f_N(\theta) + \frac{1}{2N} T_\alpha(\theta).$$

This sign convention matches the metadata stored by the implementation: `penalty = -0.5 * trace_term` and `penalized_fmin = fmin + penalty_per_observation`.

ML and FIML

For complete-data ML, ψ_i is the row-wise gradient of the normal log-likelihood contribution in the full free-parameter basis. For FIML, ψ_i is the observed-pattern score contribution for the row's observed subvector. The FIML complete-data special case is therefore pinned to the ML path by a regression test: after scale normalization, bread, meat, trace, and the explicit correction agree.

Moment-quadratic estimators

For a moment block, let

$$D_b(\theta) = \frac{\partial m_b(\theta)}{\partial \theta^\mathsf{T}}.$$

The stage-two estimating equation has the form

$$U_N(\theta) = \sum_b \frac{n_b}{N} D_b(\theta)^\mathsf{T} W_b d_b(\theta).$$

The observed bread is the derivative of this estimating equation, not only the Gauss-Newton term. In the continuous LS path this is the same observed-Hessian bread used by the misspecification-robust SE implementation. Ordinal, mixed, and ML2S paths reuse the corresponding weighted-inference observed bread.

The meat is built from complete infinitesimal-jackknife rows for the fitted moment estimator. In block notation the implementation computes rows whose crossproduct is equivalent to

$$E_\alpha = \sum_b (D_b K)^\mathsf{T} V_b^\mathsf{T} V_b (D_b K),$$

where V_b is the casewise transformed moment influence, including any estimated-weight or first-stage contribution. This is why ordinal DWLS/WLS, mixed polyserial/polychoric LS, DLS/ADF, and ML2S do not use a stage-two-only meat: their first-stage saturated-moment and weight estimation effects are already in V_b.

Estimator routing

The common RBM result reports both full-θ and reduced-α bread/meat diagnostics. Current routed estimators are:

ML	complete-data normal scores,
FIML	observed-pattern normal scores,
ULS/GLS/WLS	continuous moment IJ rows,
ordinal ULS/DWLS/WLS	polychoric/threshold IJ rows,
mixed ordinal ULS/DWLS/WLS	mixed moment IJ rows,
ML2S	FIML saturated-moment IJ rows with NT/DWLS/ADF/DLS weig

Nonlinear equality constraints remain outside the implementation because the current correction requires a fixed linear tangent matrix K over the local finite-difference stencil.